

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. We prove a weighted Harris–Doebelin theorem for the actual Sinai past kernel, derive a uniform quenched enhanced suspension invariance principle from the correctly centered Poisson martingale, construct the eigenfunction Doob nonlinear history semigroup, and obtain a normalized instantaneous–resonant–decaying memory decomposition from a continuous-time renewal resolvent with integrable vertical bounds.

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1. ROUND-TWELVE POSITIVE CLOSURE: HARRIS PAST KERNELS, QUENCHED ROUGH LIMITS, AND RENEWAL MEMORY

The complete past is the left-infinite Markov-renewal history from A3. Its conditional future kernel is constructed from the normalized Gibbs kernel, not inferred from the path LDP. The Poisson algebra, nonlinear normalization, and the causal memory sign are fixed throughout.

1.1. A weighted spectral theorem for the actual past kernel. Let $H_k = (\dots, A_{k-1}, A_k)$ and let P append one excursion according to the A3

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kernel. Put

$$W(H) = 1 + \sum_{j \geq 0} e^{-\gamma j} e^{\eta(r(A_{-j}) + t(A_{-j}))}.$$

Let $\mathcal{H}_{W,\alpha}$ be the weighted dynamically Hölder functions on pasts.

Theorem 1.1 (Past-kernel Harris–Doebelin theorem). *There are $C < \infty$, $\rho < 1$, a compact small set K , and a probability ν_K such that*

$$PW \leq \rho W + C, \quad P^m(H, \cdot) \geq \epsilon \nu_K(\cdot), \quad H \in K.$$

Consequently P is Feller on the weighted history topology and

$$\|P^n f - \pi(f)\|_{W,\alpha} \leq C \rho^n \|f\|_{W,\alpha}.$$

All constants are locally uniform under the bounded source twists used in A2 and A3.

Proof. The newest coordinate contributes the A3 exponential return moment; all older coordinates are shifted and multiplied by $e^{-\gamma}$. Choosing γ and the moment exponent gives the drift. On a finite return/homogeneity core, the connector theorem supplies one common regular word of bounded length. Bounded distortion gives a uniform lower density on its terminal cylinder, which is the minorization. Coupling at the minorization time and using the drift on excursions from K proves the weighted contraction. Continuity of the Gibbs branch weights and of the shift gives the Feller property. $\square$

1.2. Poisson martingales and the quenched suspension limit. For centered $g \in \mathcal{H}_{W,\alpha}$ define

$$h = \sum_{j \geq 0} P^j g, \quad D_{k+1} = h(H_{k+1}) - Ph(H_k).$$

Then D_{k+1} is a martingale difference and

$$g(H_k) = D_{k+1} + h(H_k) - h(H_{k+1}).$$

Lemma 1.2 (Uniform Poisson and bracket estimates). *The series for h converges in $\mathcal{H}_{W,\alpha}$. For every compact W -sublevel of initial pasts, predictable brackets of the martingale array converge locally uniformly to the Green–Kubo covariance, the conditional Lindeberg condition holds, and the coboundary divided by $\sqrt{n}$ vanishes uniformly in probability.*

Proof. Theorem 1.1 sums the Poisson series. Apply the same spectral estimate to the centered conditional second moment $P(hh^T) - (Ph)(Ph)^T$ to obtain bracket convergence. Exponential clock and observable moments imply Lindeberg by truncation. The drift inequality makes $h(H_n)/\sqrt{n}$ uniformly negligible on compact initial W -sublevels. $\square$

Theorem 1.3 (Quenched enhanced suspension invariance principle). *For every finite vector of centered dynamically Hölder observables, the collision sums and their canonical iterated integrals converge, uniformly on compact past sets, to a Brownian rough path with covariance*

$$D = \pi(DD^T)$$

and deterministic antisymmetric area correction

$$\Gamma = \frac{1}{2} \sum_{k \geq 1} \pi(g \otimes P^k g - P^k g \otimes g).$$

The same statement holds in physical time after A3 roof stopping; the terminal flight is jointly negligible at diffusive scale.

Proof. Lemma 1.2 and the martingale functional central limit theorem give stable first-level convergence. Burkholder estimates and the bracket theorem give tightness and convergence of the martingale iterated integrals. Expanding the coboundary cross terms yields the absolutely convergent area series. The A3 joint roof LDP and exponential moment imply a uniform renewal law of large numbers and $o(\sqrt{T})$ residual clock; the random-time-change theorem then gives the physical suspension result. $\square$

1.3. The genuine nonlinear history semigroup. For a bounded potential V , let T_t^V be the Feynman–Kac semigroup. Let $e^{t\lambda_V}$ and $h_V > 0$ be its leading eigen-data from Theorem 1.1. Define

$$\hat{P}_t^V F = e^{-t\lambda_V} h_V^{-1} T_t^V(h_V F), \quad \mathcal{E}_t^V F = \log \hat{P}_t^V(e^F).$$

Proposition 1.4 (Doob nonlinear tower). $\hat{P}_t^V$ is a Markov semigroup on the weighted history space and $\mathcal{E}_{s+t}^V = \mathcal{E}_s^V \mathcal{E}_t^V$. Its generator on the Doob graph core is $e^{-F} \hat{L}_V e^F$.

Proof. The eigenfunction relation cancels the Feynman–Kac growth and gives $\hat{P}_t^V 1 = 1$. The linear tower then survives conjugation by the same eigenfunction, and exponentiation gives the nonlinear tower. Differentiation on the graph core gives the generator. $\square$

1.4. Continuous-time renewal resolvent and memory. Let L_V be the closed generator of the Doob suspension in $L^2(\hat{\pi}_V)$ and let R be a finite-rank orthogonal resolved projection. Entrance, complete-excursion, and terminal-flight operators give for $\Re z > 0$

$$(z - L_V)^{-1} = E_V(z)(I - \mathcal{L}_{V,z})^{-1} F_V(z) + G_V(z).$$

Theorem 1.5 (Renewal resolvent strip). *There are $\gamma > 0$ and $\delta > 0$ such that the preceding identity extends meromorphically to $\Re z > -\gamma$. After subtracting the finitely many large- z polynomial terms and the finitely many poles/zeros in $\{|\Im z| \leq B\}$, the residual compressed transfer function and its first two vertical derivatives are $O((1 + |\Im z|)^{-1-\delta})$ on every smaller strip. The number of transmission zeros there is finite.*

Proof. The renewal identity follows by splitting a suspension path into entrance, complete returns, and the terminal flight. The A2 small/compact/moderate/ very-high estimates apply to $(I - \mathcal{L}_{V,z})^{-1}$; the entrance and flight operators gain one vertical power by integration over age. Repeating that integration twice gives the stated integrable

derivatives. High frequency invertibility follows from the multi-block estimate. Analyticity then confines poles and determinant zeros to a compact rectangle, where they are finite unless the determinant is identically zero; strict accretivity in the right half-plane excludes that alternative. $\square$

Define

$$C_V(z) = R(z - L_V)^{-1}R, \quad \widehat{K}_V(z) = zR - RL_VR - C_V(z)^{-1}.$$

Theorem 1.6 (Unique normalized memory decomposition). *There is a unique decomposition, normalized by controllable–observable minimal realization,*

$$K_V = \sum_{j=0}^J A_j \delta_0^{(j)} + \sum_{\ell=1}^N e^{\zeta_\ell t} P_\ell + K_V^{\text{reg}}(t),$$

where $\Re \zeta_\ell < 0$ and $\|K_V^{\text{reg}}(t)\| \leq Ce^{-\gamma' t}$. For every resolved observable the projected Koopman evolution satisfies the exact causal Volterra equation with this kernel. Descriptor realizations are unique up to similarity after minimality and the monic high-frequency normalization are imposed.

Proof. Polynomial large- z terms invert to the displayed instantaneous distributions. Residues of poles and transmission zeros give a finite minimal descriptor system. The residual transform and two vertical derivatives are integrable by Theorem 1.5; shifting the Bromwich contour gives the exponentially decaying function. Laplace transformation of the projected resolvent identity gives the Volterra equation. Standard controllability/observability reduction and monic normalization give uniqueness up to similarity. $\square$

Theorem 1.7 (Round-twelve A4 closure). *The Sinai past kernel has a proved weighted spectral gap, a uniform quenched rough suspension limit, an exact Doob nonlinear history semigroup, and the normalized distributional-plus-decaying memory decomposition of Theorem 1.6.*

Proof. Combine Theorems 1.1, 1.3, 1.5, and 1.6, together with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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